

The whole BMS below FMP(1,2,2,4,2,4,4,4)

DEFINITIONS & PROOFS / ENGLISH EDITION

2026-09-23. A paper comparison for the **original FMP fundamental sequence**: copy n times, delete the final self-copy, then fully complete. See the definition, kernel, and well-ordering manuscript.

Status: author-audited paper proof, not an independent referee report or a Lean theorem. The ordinal interpretation uses the companion FMP well-ordering result in **ZFC + I3**. The combinatorial simulation below introduces no additional large-cardinal hypothesis. The bounded test is an implementation audit, not a replacement for the argument.

1. Statement and conventions

Let Θ_{BMS} be the order type of all finite standard ordinary BMS expressions, excluding an adjoined display top. Put

$$H = [][1][1][1:3^*;2][1][1:5^*;2][1:5^*,2:6^*;3][1:5^*,2:6^*;3].$$

In FMP's compact syntax, the final number in a column is its pivot; the two endpoint edges are implicit. The exact count word of H is 1, 2, 2, 4, 2, 4, 4, 4.

Theorem. The whole standard BMS order strictly order-embeds into the strict standard FMP lower segment of H . Consequently,

$$\Theta_{\text{BMS}} \leq o_{\text{FMP}}(H). \quad (1)$$

Here $o_{\text{FMP}}(H)$ is the order type of the expressions strictly below H . “Strictly order-embeds” means preservation of strict comparisons; it does **not** turn (1) into a strict ordinal inequality.

This is not an equality theorem, a minimal-carrier theorem, or a termwise identification of the two fundamental sequences. In particular, no upper bound $o_{\text{FMP}}(H) \leq \Theta_{\text{BMS}}$ is established.

All target addresses are one-based; BMS columns and rows are zero-based. Truncation to a shorter prefix always denotes finitely many actual FMP $[0]$ steps. No completion, mark erasure, or arbitrary insertion is an extra descent operation.

2. The source BMS facts

Write $B_h = (0^h)(1^h)$, for $h \geq 1$, and use the short basic sequence: index zero deletes the last column, and index $n > 0$ appends n copies of the bad block. Trailing zero rows are ignored. Then $B_{h+1}[1] = B_h$; their descendant sets are nested and exhaust the standard domain.

We use the ordinary BMS parent-table lemma, in the precise form already documented in the collected BMS-to-ACD proof, Section 2. That section records its Hunter Lemma 2.5 provenance and pinned Lean declarations; this FMP addition does not rebuild or extend those Lean results.

Let x be the last source column, r its highest nonzero row, $c = P_r(x)$, and $L = x - c$. Set

$$\phi_t(p) = \begin{cases} p, & p < c, \\ p + tL, & p \geq c. \end{cases}$$

An absent parent stays absent. The prefix $j < x$ is unchanged, and for $1 \leq t \leq n$ the copied parent table satisfies

$$P'_q(j + tL) = \phi_t(P_q(j)) \quad (c < j < x), \quad (2)$$

$$P'_q(c + tL) = \begin{cases} \phi_{t-1}(P_q(x)), & q < r, \\ P_q(c), & q \geq r. \end{cases} \quad (3)$$

For $q < r$, $P_q(x)$ exists and is at least c . Standard numerical entries are the corresponding row-parent-chain depths. Thus this parent-table description specifies the actual BMS expansion, not a separate arbitrary-graph system.

We also use the standard prefix-barrier property: for two standard matrices $M < N$, M is obtainable from N by a nonempty finite sequence of basic-sequence steps. One way to see this is to place their two finite histories below a common B_h . At the first differing basic index, the earlier child is a prefix of the later one. Subsequent steps preserve that prefix until they reach it; all prefixes are obtainable by deletion. If a branch passes it, its first crossing must equal it. Iterating this observation down the finite histories gives the claim. The reverse implication follows from the usual strict decrease of BMS expansion in column order.

No statement here concerns every syntactically valid nonstandard matrix.

3. A standard eight-column carrier

Let $A = S_1[0][1] = \square[1][2]$. Its second basic term has count word 1, 2, 2, 4, 2, 4, 4, 8. Starting there, perform four rounds of:

1. Take the actual FMP term at index 1.
2. Delete last columns until exactly eight remain.

The successive last counts are 7, 6, 5, 4; the resulting full marked graph is exactly H . This is a fixed finite substitution in the definition, not an argument that every graph with these counts is standard. It proves the required standard reachability.

Write $V = H[0]$. For a source base u and a target base v , define the port at column $v + q$ by

$$P_q(u, v) = \{u + i \mapsto v + i : 0 \leq i \leq q + 1\}, \quad (4)$$

with marked targets $v, \dots, v + q - 1$. For $q = 0$ there are no marks.

Set $D_0 = V$. For $j \geq 1$, put

$$g_j = j + 2, \quad u_j = j^2 + 3j + 1, \quad c_j = u_j + g_j, \quad b_j = u_j + 2g_j.$$

Append the following two bands to D_{j-1} :

$$P_0(u_j, c_j), \dots, P_{g_j-1}(u_j, c_j),$$

$$P_0(u_j, b_j), \dots, P_{g_j}(u_j, b_j).$$

The exact basic-term formula is

$$\boxed{H[m] = D_m, \quad |D_m| = m^2 + 6m + 7.} \quad (5)$$

This uses FMP's real copy index m , not an index chosen afterwards from the output length.

Direct proof of the full-completion formula

Here are explicit raw tables, to avoid importing any deferred-completion or repeated-sweep equivalence.

Write $C(u, v, e)$ for the three-edge column at $v + 1$ with edges

$$(u, v), (u + 1, v + 1), (e, v + 2),$$

marking target v . Write $R(u, b, c)$ for the column at $b + 2$ with edges

$$(u, b), (u + 1, b + 1), (c, b + 2), (c + 1, b + 3),$$

marking targets $b, b + 1$.

After m raw COPY operations and deletion of the actual final self-copy, the graph is V followed, for each $1 \leq j \leq m$, by

$$P_0(u, c), C(u, c, c), P_0(u, b), C(u, b, c), R(u, b, c), \quad (6)$$

where $u = 5j, c = 5j + 3, b = 5j + 5$.

For the first copy, the controller sends $1 \mapsto 5, 2 \mapsto 6$, and every $x \geq 3$ to $x + 5$; its six source columns give (6) and the new controller. At the next round the latter has the corresponding shifted table. Its possibly unavailable intermediate point occurs in none of the copied tables. The marked traces being copied are all in the high part, so the stated marks survive. This proves (6) by induction.

Now scan (6) using the actual full-completion rule. At stage j , let $g = g_j$ and let α denote the final address of an old point. Induction over the five entry-old columns gives

$$\begin{aligned} \alpha(u) &= u_j, & \alpha(u + 1) &= u_j + 1, \\ \alpha(c) &= c_j, & \alpha(c + 1) &= c_j + 1, \\ \alpha(b) &= b_j, & \alpha(b + 1) &= b_j + 1, \\ \alpha(b + 2) &= b_j + g, & \alpha(b + 3) &= b_j + g + 1. \end{aligned} \quad (7)$$

The first and third columns have native width zero. Each C has width

$$\alpha(c) - \alpha(u + 1) - 1 = g - 2;$$

its restrictions supply ports $1, \dots, g - 1$ after the preceding port zero. The old marked edge of each C has a width-zero bottom and supplies no extra packet.

The last column R has native width zero. Its first old mark again has bottom width zero. Its second mark has bottom at the second C , of width $g - 2$, and therefore supplies exactly the missing edges

$$u_j + i \mapsto b_j + i \quad (2 \leq i \leq g - 1).$$

Together with its two endpoint edges, this is exactly $P_g(u_j, b_j)$, including all marks. The five raw columns thus become the two bands in (5). The next source base is this stage's second target base, and the next band starts after its final port; these recurrences give u_j, g_j, c_j, b_j above. Summing the $2g_j + 1$ new columns proves the width formula.

This is an induction over the original one-pass COMPLETE events. No auxiliary scan has been treated as a descent.

4. Preparing arbitrarily many BMS rows

Fix $m \geq 1$, abbreviate $g = m + 2, u = u_m, b = u + 2g$, and set

$$Z_m = D_m[0], \quad E_m = Z_m[1], \quad h = g - 1, \quad s = 2g, \quad \ell = b - 1. \quad (8)$$

In E_m , choose bases b and $b + s$, and take the first h ports at each. Their parents encode precisely B_h .

For completeness, the entire suffix of E_m starting at b is as follows. Put $v = 1$ if $m = 1$, and $v = u_{m-1}$ otherwise:

$$P_0(u, b), \dots, P_{g-2}(u, b), P_{g-1}(v, b),$$

$$P_0(b, b + g), \dots, P_{g-1}(b, b + g),$$

$$P_0(b, b + 2g), \dots, P_{g-2}(b, b + 2g). \quad (9)$$

The prefix before b is unchanged. Substitute the controller $P_{g-1}(u, b)$ into COPY: its address map fixes points below u and shifts every point at least u by $2g$. The first copied source column is the last port of D_{m-1} ; this gives the exceptional filler $P_{g-1}(v, b)$. The remaining columns give the two later bands in (9). All final source pairs are adjacent, so COMPLETE is inactive. Marks are retained according to the actual rule, not assigned by hand.

The first selected band has external source u , with $u + h < b$; the second has source b . Every active cap source avoids both selected bases. The prefix of length ℓ strictly extends D_{m-1} . Hence all representations preserving this prefix and descending from E_m lie in

$$(D_{m-1}, D_m). \quad (10)$$

This preparation costs two genuine steps from D_m .

5. The closed band simulation

Fix h, s, ℓ as above. A representation of a k -column standard matrix has selected bases

$$v_j = \ell + 1 + sj \quad (0 \leq j < k)$$

and ends at column $v_{k-1} + h - 1$; the empty matrix is represented by the protected prefix of length ℓ . Require:

1. At $v_j + g$, for $0 \leq q < h$, the unmarked table has the form $P_q(u, v_j)$.
2. If the BMS parent is i , then $u = v_i$. If there is no parent, $u + h < v_0$.
3. Every nonempty active column, including fillers, is saturated, and its cap source is not a selected base.
4. The entire protected prefix, including its marks, is fixed. All other marks are those obtained by actual expansions.

Equation (9) satisfies these conditions. Marks on later representations need not be canonical.

For a source step at index zero, or from a matrix whose last column is zero, simply delete the last selected band and the preceding filler columns. This is a nonempty finite string of $[0]$ operations.

Otherwise let r be the highest active source row, $c = P_r(k - 1)$, $v = v_{k-1}$ and $u = v_c$.

1. Truncate to the controller at $v + r$.
2. Apply one actual FMP $[n]$.
3. Truncate to the last required complete selected band.

The controller's table (4) makes its address map **total**:

$$\psi(x) = \begin{cases} x, & x < u, \\ x + v - u, & x \geq u. \end{cases} \quad (11)$$

This includes $r = 0$, without an exceptional rule.

Why full completion is inactive

An adjacent source pair can acquire a gap under (11) only if its cap source equals u . Condition 3 excludes this, since u is a selected base. Thus every copied column stays saturated; no owner acquires positive completion width, and no marked packet is propagated. Empty maps stay empty. We still execute the actual star-transfer rule, but its result cannot change this saturation argument.

The first copied fragment fills the missing rows of the current last band; later fragments give regularly spaced new bands. Their bases are

$$(v_0, \dots, v_{k-2}) \text{ followed by } v_j + t(v - u) \quad (1 \leq t \leq n, c \leq j < k - 1). \quad (12)$$

At the first new base, rows below r were retained from the controller band, and rows at least r come from the copied bad-root band. This is exactly (3). For all other bands, source and target bases are transported together, giving (2).

There is no hidden split within a selected port: an internal source base is separated from other bases by $s \geq h + 1$, while an external source interval lies entirely below v_0 . Thus its consecutive sources lie wholly on one side of the cut in (11). Cap avoidance is preserved because (11) is injective and no old non-base cap can acquire the image of an old base; unchanged low caps remain low. This proves all four representation invariants again, including the width and full-prefix conditions.

Every source step is consequently simulated by a nonempty finite sequence of actual target steps, from **every** representation in this closed sector. This is a branching simulation of the full descendant cone, not just one preferred path.

6. From simulations to one order embedding

A choice of a simulation history could give several marked graphs for the same matrix. We must remove that ambiguity before claiming an order embedding.

For fixed m and $M \in \text{Desc}(B_{m+1})$, let $\mathcal{R}_m(M)$ consist of all representations reached from E_m by matching finite source histories. It is nonempty and closed under continuing any source step. Every member has the same width:

$$\ell + s(k - 1) + h \quad (k = |M| > 0), \quad \ell \quad (k = 0). \quad (13)$$

There are only finitely many marked FMP graphs of a fixed width: addresses are bounded by the width plus one, maps and star sets are finite subsets of fixed finite sets. Thus $\mathcal{R}_m(M)$ is finite. Take its least member in the standard FMP order:

$$e_m(M) = \min \mathcal{R}_m(M).$$

The companion well-ordering manuscript supplies the standard total order and strict descent of every actual step. In fact this minimization needs only a finite total order, not an additional global well-ordering argument.

If $M < N$ in the same cone, Section 2 gives a nonempty finite source path from N to M . Simulate it starting from $e_m(N)$, obtaining some $G \in \mathcal{R}_m(M)$. Then

$$e_m(M) \leq G < e_m(N). \quad (14)$$

All these images lie in (10).

Finally assign each standard BMS matrix M the least integer $m \geq 1$ with $M \leq B_{m+1}$, and use $e_m(M)$. The assignment is defined and nondecreasing because the seeds are cofinal and nested. If $M < N$ have the same assigned integer, apply (14). If their assigned integers differ, the disjoint ordered intervals (10) give the strict comparison. This produces **one** embedding of the entire BMS order into the strict lower segment of H , proving (1).

Neither variable row capacity by itself nor an enumeration of finite sample paths would have sufficed for this conclusion.

7. Reproducibility and limits

Run the self-contained comparison audit:

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python -B tests/fmp_bms_comparison.py
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It checks standard reachability of H , the raw five-column formula and completed formula for $m = 1, \dots, 12$, all selected parents and recovered numerical depths, the protected prefixes, and the absence of any extra completion in the simulation. Its deterministic descent includes every active row down to zero and terminal deletions. Random branches use the numerical BMS rule, independently of the FMP parent transport.

The run is bounded by 20 seconds, 2,000 random attempts, 1,600 columns and 112 retained states; width skips are reported, not counted as successes. The recorded release run passed 114 directed macros and 1,742 random macros, with peak width 1,297. See VALIDATION for release checks.

The proof uses all-parameter inductions and the existing BMS parent lemma. These tests do not certify it formally. No new FMP Lean module or comparison certificate is claimed. Equality with BMS, strictness above BMS, and a smaller FMP carrier remain open here.